

A scale-free proof of the connector seam sign at Poincaré's point D

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Abstract

The last unresolved numerical issue in the current connector construction is caused by a double zero at the collision point D . Direct interval evaluation loses the small common Morse scale and produces boxes containing zero. We retain that scale algebraically. The oriented real derivative is written exactly as

$$L \operatorname{Re} X_s(L, \delta) + \delta^2 \operatorname{Re} Z_s(L, \delta).$$

Here $s = -1$ and $s = +1$ denote the initial and final connectors. An independent outward-rounded Python interval calculation proves $\operatorname{Re} X_s \geq 0.00419$ and $\operatorname{Re} Z_s \geq 0.06558$ on both complete collars. Since $L > 0$, the required derivative orientation follows uniformly, however small the nonconstructively selected Morse length is. Ordinary one-variable calculus then supplies the missing connector seam sign.

The remaining 27/2744 radial gap is now closed as well. A 40-bit centered Taylor table certifies strict decrease on 14,336 continuum cells. A separate endpoint certificate uses direct positivity away from the pinch and exact double-zero anchors plus positive second derivative on the collision collars. Lean reconstructs both tables semantically and obtains an unconditional principal square-root sheet on the complete pre-collision standard circle.

1 Scope

This note addresses the local-to-global seam problem for the two affine connectors at the specific collision point D . It uses analytic facts already established in the project: the inverse-Morse endpoint can be chosen with the normalized bounds in §5, the companion collision factor is positive on the collars, and the first factor has the required imaginary monotonicity. The new content is the real derivative sign for the first factor.

The subsequent Lean development reuses the same bulk rows uniformly in the critical-value coordinate and joins them to this scale-sensitive collar argument on every positive fiber. It therefore proves connector-radical nonvanishing and constructs a continuous square-root sheet on each half-open rectangle $[0, 1) \times [0, 1]$. The later file `ChapterVIDJointLiftedContour.lean` assembles those sheets with the outer and middle sheets into one closed, basepoint-normalized lifted five-piece family near D . A subsequent 44-shard certificate proves nonvanishing on the complete moving angular circle in the transformed root coordinate through the exact cutoff 2717/2744, and constructs its principal lift. The initial source-circle identification is also closed: on the unit circle the exact coordinate map is the orientation-preserving angular reparametrization $\theta \mapsto \theta - (200/30003) \sin(3\theta)$, and Lean transports the principal sheet through its continuous inverse. The centered derivative and endpoint tables described below now close the final 27/2744 too. What remains is not a numerical radial gap: it is the corrected global jointly moving source-cycle transport, its deformation into the terminal five-piece cycle, and the derivation of every source-specific multiplier, interchange, and coefficient-identification statement in §102 from the full Chapter V uniform-integral problem.

2 Map to Poincaré's printed proof

The following table distinguishes three different situations: an actual algebraic or typographical error in the printed formulas; a substantial argument that Poincaré explicitly sketches but does not prove; and a modern analytic hypothesis that is needed to make his intended inference rigorous. Calling all three an “error” would be misleading. The third column therefore states the precise defect, and labels genuine formula errors explicitly.

source	printed role	error, omission, or rigor issue	what this project adds and present status
§§90–92	Introduce the perturbing Hamiltonian, osculating variables, and the first homological equation.	No isolated formula error is asserted here. The issue is scope: the current theorem models the circular restricted principal term, not every coordinate and uniform first-integral hypothesis of the full three-body discussion.	The repository formalizes the restricted Hamiltonian derivative, product-rule obstruction, and coordinate-independent algebra. A faithful full-source theorem would still need the exact Chapter V–VI coordinate maps and uniform-integral definition.
§93	Use Darboux’s principle to read coefficient asymptotics from boundary logarithms.	The printed application suppresses the uniform analytic estimates needed when amplitudes and parameters vary. A local logarithm alone does not imply that the remaining factor is analytic on a larger common disk.	Formal coefficient extraction, equal-modulus separation, finite-jet and Tannery variants, and the larger-disk regular-factor transfer are proved. Their source-specific uniform hypotheses remain to be derived from the actual contour integral.
§94	Form the one-variable Laurent function $\Phi(z)$ by integrating a Fourier ray around the unit circle.	Infinite-series convergence, interchange of sum and contour integral, and branch choices are not fully exposed in the printed argument.	Lean defines the literal normalized contour integral, proves the finite and absolutely summable affine-ray coefficient extraction, and constructs the explicit unit-circle path. The complete physical Fourier series and its common annulus are still source inputs.
§95	Locate possible singularities when moving poles obstruct contour deformation.	Solving the discriminant equations finds candidates but does not show that the original cycle is actually pinched. Genuine versus apparent singularities require sheet and cycle data.	The project proves closed-contour homotopy invariance, winding obstructions, exact cycle splitting, and a local vanishing-cycle theorem. Placement of the actual source-sheet cycle into the certified contour remains global work.
§96, p. 290	Derive algebraic collision equations using half-angle variables.	Printed formula error: equation (10) contains $x^2 - 1$, while the preceding equation and the stated reciprocity require $x^2 + 1$.	The corrected reciprocal equation is derived and formalized; the resulting collision cubics, source radicand, concrete candidate D , and fiber double-zero facts are checked exactly.

source	printed role	error, omission, or rigor issue	what this project adds and present status
§§97–98	Decide which candidates are admissible and continue the integration cycle on the Riemann surface.	Poincaré says at the end of §98 that the discussion is only sketched and that a complete study of the branches of Φ is needed. The printed text does not construct the global sheet-preserving deformation.	Lean proves the concrete root branches, synchronized real parameter, and proves that Poincaré’s exact $u \mapsto t$ map carries the u -unit circle to the literal §94 t -unit circle through an explicit homotopy of parametrizations at a fixed parameter. It also proves that this fixed circle cannot be the continued cycle near D : the pole that begins outside has crossed strictly inside, and winding-number invariance forbids a pole-avoiding fixed-circle continuation. Lean further proves the outer arcs, affine connectors, and the complete seam tables. The old bulk rows are now used uniformly in the critical-value coordinate, the homogeneous theorem covers every positive-fiber collar, and covering-space lifting constructs both connector sheets on $[0, 1) \times [0, 1]$. Lean also proves that each transformed connector integral is literally the corresponding source curve integral. The terminal construction now concatenates the two outer quarters, both connectors, and the inverse-Morse middle path into one closed moving source cycle. The literal radicand is nonzero on the positive half-open family, one basepoint-normalized lift is constructed, and its five restrictions are proved equal to the certified branches. The long radial certificate first used 1,408 polar cells in 44 LeanCompCert shards through $s = 2717/2744$. The repaired centered certificate now proves strict radial decrease on the final $27/2744$, while a direct/convex endpoint certificate supplies the collision-circle nonnegativity. Together they prove positive real part and a principal lift on the ordinary angular root-coordinate circle for every $s < 1$. The rational quarter parametrization is proved surjective onto each geometric quadrant. The remaining geometric obligation is the corrected jointly moving source-cycle and sheet transport into the near- D five-piece family; equality of norms does not prove that this source-coordinate homotopy avoids the moving zeros.

source	printed role	error, omission, or rigor issue	what this project adds and present status
§99	Move the double zero to a critical center and reduce the radicand to a quadratic pinch.	Formal Weierstrass manipulation by itself would not establish convergence, a jointly analytic coordinate change, its Jacobian, or a nonzero pulled-back numerator.	Complex implicit-function, Hadamard-division, analytic Morse-coordinate, Jacobian one-form, transversality, literal numerator, and compact local vanishing-cycle theorems are proved for D .
§99, p. 323	State the leading pulled-back amplitude $\theta_{0,0}$.	Printed formula error: after $\Delta = \frac{1}{2}\Delta_{tt}(t - t_0)^2 + \dots$, the printed factor multiplies by $\frac{1}{2}\Delta_{tt}$; the preceding defining identity instead gives a factor proportional to $\sqrt{2/\Delta_{tt}}$, up to sheet sign.	The project derives the amplitude from the actual inverse Morse coordinate and Jacobian. Its exact value has the reciprocal-square-root behavior required by the defining identity, rather than assuming the inconsistent printed expression.
§100	Integrate the prepared inverse square root to obtain $\Phi_2 + \Phi_3 \log(z - z_0)$, then apply Darboux.	The local primitive is plausible, but the passage to the global source contour and the larger-common-disk control of varying analytic amplitudes are omitted. Nonconstant log amplitudes cannot simply be absorbed into an analytic remainder.	Lean proves the function-level logarithmic primitive, straight-arc integral, nonzero local coefficient, three- and five-piece asymptotic interfaces, and the required Darboux transfer under explicit uniform hypotheses. The radial numerical gap is now closed; the source-cycle deformation and regular remainder are still needed.
§101	Give the Pallas numerical example.	No defect relevant to the logical nonintegrability argument is asserted; this is an application rather than a required proof step.	Not formalized, and not needed for the present theorem.

source	printed role	error, omission, or rigor issue	what this project adds and present status
§102	Infer that a uniform integral forces the singular-point motion to depend on too few parameters.	The reference to Chapter V no. 85 contains a printed normalization error . With $D_\lambda = B_\lambda \zeta^\lambda$, equation (13) gives $dB_\lambda(X_\Phi) = \lambda B_\lambda S$. The printed relation $-\zeta H = S$, or $H = -S/\zeta$, does not make $dD_\lambda(H, X_\Phi)$ vanish; the required relation is $H = -\zeta S$. The printed argument also does not supply the full analytic identification of these coefficients with the contour singularities.	Lean formalizes Fourier integration on the actual two-dimensional unit additive torus. From one pointwise first-order Poisson identity, linearity gives the head and five tail coefficient equations; the six equations are not separately assumed. The torus model then uses the standard angle-derivative multiplier and orbital differentiation-under-the-integral rules, resonance (12 bis), and independence to derive equation (13), the corrected cancellation, and the complete rank chain. An exact rational instance $(\lambda, \zeta, B, S) = (1, 2, 1, 1)$ gives residual 3/2 for the printed normalization. With the correction, a common nonzero characteristic direction gives rank at most five; quotienting the common scale gives rank at most four; independent recovery of the eccentricities gives rank at most two; and §103 contradicts this. It remains to derive the multiplier, interchange, and independence fields from the full source Hamiltonian and uniform integral, and identify the six observables with the continued-contour singularity data, including the common radius and regular-factor estimates.
§103, p. 331	Differentiate $P = \sum U^2$ after defining $V = x \partial U / \partial x - U$.	Printed formula error: the text gives $xP_x = 2 \sum VU + P$; substitution gives $xP_x = 2 \sum VU + 2P$. Since the next step restricts to $P = 0$, this typo does not change the reduced equation.	Lean verifies both the corrected global identity and its on-curve specialization. It also proves the concrete sextic/septic geometry, 24-point locus, local multiplicities, moving analytic branches, differential identity, and final finite rank contradiction.

The contribution of the present note is concentrated in the §§97–100 rows: it closes the scale-sensitive connector sign and the former radial-tail numerical gap. It deliberately does not use those local successes to conceal the still-open source-cycle and §102 hypotheses.

3 The certified global prefix

The long-range certificate uses the elementary quarter-circle parametrization

$$Q(t) = \frac{1 - t^2 + 2it}{1 + t^2}, \quad 0 \leq t \leq 1,$$

and its rotations Q , iQ , $-Q$, and $-iQ$. Lean proves that these four maps cover the ordinary root-coordinate unit circle: in the first quadrant the inverse parameter is $t = \operatorname{Im} z / (1 + \operatorname{Re} z)$, and the other quadrants reduce to this one by exact rotations. Hence a certificate evaluated using Q applies pointwise to the standard angular parametrization, with no unproved reparametrization assumption.

item	exact value	kernel-checked conclusion
radial mesh	28 cubic-clustered cells	first 22 rows used on pinching half
angular mesh	32 quadratic-clustered cells per quarter	continuum cover, not point samples
new artifacts	$2 \cdot 22 = 44$	independent verified-program verdicts
new polar cells	$2 \cdot 22 \cdot 32 = 1408$	strictly positive real radicand
terminal prefix	$1 - (6/28)^3 = 2717/2744$	one principal sheet on the full root circle
repaired tail	$1 - 2717/2744 = 27/2744$	strict radial decrease plus endpoint nonnegativity

The fixed cutoff by itself did not prove overlap with the analytic Morse neighborhood: the local inverse-function argument supplies a positive neighborhood but no numerical lower bound for its size. Merely moving the cutoff closer to one would not have removed that logical issue. The repair instead proves a global monotonicity statement all the way to the collision endpoint, so it does not need a quantitative size for the Morse neighborhood.

There is, however, a scale-free monotonicity reduction. Write $R(s, h)$ for the radicand on either left quarter, where h is angular distance from the negative-real pinch. It is enough to certify

$$\Re R(1, h) \geq 0, \quad -\partial_s \Re R(s, h) > 0 \quad (2717/2744 \leq s < 1).$$

The mean-value theorem then gives $\Re R(s, h) > 0$ for every pre-collision point. The first quantity has a genuine quadratic zero at $h = 0$. The implemented endpoint proof uses direct positivity away from that zero, and on a small collar proves the value and first derivative vanish exactly and the second derivative is positive. Convexity then gives the nonnegative endpoint value without dividing by a vanishing interval. The following exploratory values selected the mesh; they are not proof certificates.

row	lower s	upper s	sampled minimum of $-\partial_s \Re R$
22	0.9901603499	0.9943057580	4.5709
23	0.9943057580	0.9970845481	12.1834
24	0.9970845481	0.9987700437	36.6050
25	0.9987700437	0.9996355685	118.5914
26	0.9996355685	0.9999544461	279.1160
27	0.9999544461	1.0000000000	398.2389
endpoint	$h = 0$	$h = \pi/2$	$\min \Re R(1, h)/h^2 \approx 3.40738$

`ChapterVIDRadialTailReduction.lean` proves the exact calculus implication and its combination with the prefix theorem. The new files `ChapterVIDRadialTailDerivative.lean` and `ChapterVIDRadialTailPathDerivative.lean` prove the exact total chain rule, including the motion of both $\zeta(s)$ and $u(s)$, and specialize it to the affine parameter and sixth-root radius. They also prove that a negative derivative table supplies the semantic strict-antitone hypothesis by the mean-value theorem. The product rule is expanded first so that its two exact $4L$ contributions cancel symbolically.

The first signed-dyadic evaluator and six-row cover were encoded in `ChapterVILeanCompCertRadialTailDeriv.lean` and `ChapterVIDRadialTailDerivativeCompiledGrid.lean`. That direct 20-bit layout failed for a mathematically meaningful reason: independent boxes leave intermediate terms of size hundreds while the true first-row margin is about 4.57. A degree-five anomaly trace proves

$$\left| e^a - \sum_{k=0}^5 \frac{a^k}{k!} \right| \leq \frac{|a|^6}{512}$$

from Mathlib’s complex exponential bound and checks every polynomial operation and error radius with signed dyadics. Substituting that trace leaves the first upper-row bound near $3.94 \cdot 10^8/2^{20}$; even making the angular box a point leaves approximately $[-3.02 \cdot 10^8, 2.94 \cdot 10^8]/2^{20}$ on the first radial cell. Therefore the controlling loss is radial dependency among large Laurent terms, not the exponential remainder.

The completed computation uses the required dependency-preserving repair. The files `ChapterVIDRadialTailB.lean` and `ChapterVIDRadialTailCenteredExpression.lean` expand the expression at each radial-cell center, carry radial and angular affine coefficients separately, and bound the remaining Taylor error. The generic semantics in `ChapterVIFieldExpressionTrace.lean` prove that a valid trace encloses the literal expression and that a jet trace encloses its directional derivative. At denominator 2^{40} , kernel-evaluated integer comparisons verify every whole cell and prove that output register 52 is the actual radial derivative with negative real upper endpoint.

The endpoint uses a second, deliberately nonuniform proof. Direct positivity handles cells with a strict margin. On the upper collar $[961/1024, 1]$, exactly 63 cells of width $1/1024$ bound the second derivative from below; the lower collar is the single cell $[0, 1/1024]$. Lean proves the endpoint value and first derivative vanish at the collision and uses convexity to fill both collars. This also repairs an off-by-one draft in which 64 upper collar cells would have extended beyond the unit interval to $1025/1024$.

certificate component	checked cells	exact formal conclusion
upper centered derivative	$6 \cdot 16 \cdot 64 = 6144$	$\partial_s \Re R < 0$ on every upper-tail cell
lower centered derivative	$2 \cdot 16 \cdot 128 + 4 \cdot 16 \cdot 64 = 8192$	$\partial_s \Re R < 0$ on every lower-tail cell
upper endpoint, direct	62	$\Re R(1, h) > 0$ through 961/1024
lower endpoint, direct	124	$\Re R(1, h) > 0$ from 1/1024 onward
upper endpoint, curvature	63	$\partial_h^2 \Re R > 0$ on [961/1024, 1]
lower endpoint, curvature	1	$\partial_h^2 \Re R > 0$ on $[0, 1/1024]$
centered derivative total	14,336	strict radial antitonicity on the final 27/2744
endpoint total	186 direct + 64 curvature	collision-circle nonnegativity

The least curvature lower numerators actually evaluated are $9912363499575/2^{40}$ on the upper collar and $43455431609765/2^{40}$ on the lower collar. These numbers are not trusted annotations: the corresponding `decide +kernel` theorems reduce the Boolean interval checks in Lean, and the trace-soundness theorems reconstruct their real-analysis meaning. No CSV, floating-point sample, external execution receipt, or admitted computation enters the final theorem.

The large centered check is split into 896 independent 16-angle chunks, with separate program-validity, derivative-sign, and Taylor-radius theorems. The 2,688 computational modules and 192 lightweight radial aggregators keep kernel reduction bounded and parallelizable. The endpoint calculation is similarly split into 256 fixed-index modules. This sharding is only a resource-control device: the aggregate theorems quantify over the original complete cell sets.

Finally, `ChapterVIDRadialTailFinal.lean` combines these results into an unconditional `ChapterVIDRadialTa`. The prefix and tail therefore give $\Re R(s, h) > 0$ for every $s < 1$ on the complete standard angular circle and construct the canonical continuous principal square-root sheet.

4 The centered collision factor

Let x_D be the unique root in $[-0.027, -0.026]$ of

$$2500x^3 + 500025x^2 + 12501x - 25 = 0,$$

let $D < 0$ be its real cube root, and put

$$y_D = \frac{(x_D - 1/100)^2}{2(1 + (1/100)^2)x_D}.$$

Write $F(\zeta, u)$ for the collision factor which vanishes twice at (ζ_D, D) , and write $G = \partial F / \partial u$. Direct algebra, with every inverse-power difference cancelled before evaluation, gives

$$G(\zeta, u) = A(u - D) + (u - D)^2 K(u) + \left(\frac{\zeta}{\zeta_D} - 1 \right) P(u), \quad (1)$$

where

$$A = \frac{\partial^2 F}{\partial u^2}(\zeta_D, D) \in \mathbb{R}, \quad A < 0.$$

The explicit formulas used by the calculation are in Appendix A.

Set

$$E(a) = \begin{cases} 0, & a = 0, \\ \frac{e^a - 1 - a}{a^2}, & a \neq 0. \end{cases}$$

Then

$$e^a = 1 + a + a^2 E(a), \quad |a| \leq 1 \implies |E(a)| \leq 1. \quad (2)$$

The nonstandard value at $a = 0$ is harmless because it is multiplied by a^2 .

5 Scale-free connector coordinates

Let $s = -1$ on the initial connector and $s = +1$ on the final connector. Let $L > 0$ be the selected Morse half-length and let δ be distance from the local endpoint:

$$\delta = 1 - t \quad (s = -1), \quad \delta = t \quad (s = +1).$$

On both collars, $0 \leq \delta \leq 261/1024$. Normalize the moving inputs by

$$u_{\text{loc},s} - D = sLq_s, \quad u_{\text{out},s}(\kappa) - u_{\text{out},s}(0) = L^2 r_s, \quad \frac{\zeta}{\zeta_D} - 1 = L^2 p.$$

The inverse-Morse model can be shrunk so that

$$0 < L \leq 2^{-20}, \quad |\operatorname{Re} r_s|, |\operatorname{Im} r_s|, |\operatorname{Re} p|, |\operatorname{Im} p| \leq 1, \quad (3)$$

and

$$-\frac{1024}{2^{20}} \leq \operatorname{Re} q_s \leq \frac{1024}{2^{20}}, \quad -\frac{158311}{2^{20}} \leq \operatorname{Im} q_s \leq -\frac{130048}{2^{20}}. \quad (4)$$

This last box is the quantitative form of the downward inverse-Morse tangent.

At collapsed scale the local-to-outer directions are

$$B_- = -D(1 + i), \quad B_+ = -D(1 - i).$$

Define the actual direction and coordinate by

$$V_s = B_s + L^2 r_s - sLq_s, \quad u - D = sLq_s + \delta V_s. \quad (5)$$

These are identities. In particular, all occurrences of the small endpoint displacement share the same L .

6 The homogeneous real-part identity

Substitute (5) and the normalized parameter displacement into (1). Define

$$\begin{aligned} X_s &= Asq_s V_s + \delta A(Lr_s - sq_s)(V_s + B_s) \\ &\quad + (Lq_s^2 + 2s\delta q_s V_s)K(u)V_s + LpP(u)V_s, \end{aligned} \quad (6)$$

$$Z_s = V_s^3 K(u). \quad (7)$$

Lemma 1 (Exact cancellation). *For either connector,*

$$\operatorname{Re}(G(\zeta, u)V_s) = L \operatorname{Re} X_s + \delta^2 \operatorname{Re} Z_s. \quad (8)$$

Proof. Expand the left side using (1) and $u - D = sLq_s + \delta V_s$. Also write $V_s = B_s + L(Lr_s - sq_s)$. Collect every term containing L into LX_s , and collect the quadratic-distance term into $\delta^2 Z_s$. The only apparent term left over is δAB_s^2 . Since A is real and

$$B_-^2 = 2iD^2, \quad B_+^2 = -2iD^2,$$

its real part is exactly zero. □

Neither coefficient in (8) is divided by L . The first term retains a positive factor L , while the collapsed part begins at order δ^2 . Thus the coefficient boxes can remain separated from zero as $L \rightarrow 0^+$.

7 Finite interval lemma

The primitive dyadic boxes, all at denominator 2^{20} , are

$$\begin{aligned} D &\in [-314053, -314047]2^{-20}, & y_D &\in [-28312, -25000]2^{-20}, \\ \frac{100}{30003} &\in [3494, 3495]2^{-20}, & \frac{200}{10001} &\in [20969, 20970]2^{-20}, \\ \frac{1}{10001} &\in [104, 105]2^{-20}. \end{aligned}$$

The cubic is strictly decreasing on $[-0.026865395705, -0.026865395704]$, takes opposite signs at its endpoints, and its negative real cube root lies in the displayed D -box. Substitution in the formula for y_D gives the displayed y_D -box.

Cover the distance interval by the 160 uniform cells

$$I_j = [1671j/2^{20}, 1671(j+1)/2^{20}], \quad 0 \leq j < 160.$$

Indeed, $160 \cdot 1671/2^{20} = 0.254974365234375 > 261/1024$. On each cell and side, rectangular complex interval arithmetic evaluates (5), the Appendix formulas, and (6)–(7). The exponential remainder $E(a)$ is enclosed by $[-1, 1] + i[-1, 1]$, using (2); the same interval evaluation first checks $|a| \leq 0.218249762481 < 1$ on every cell. All other normalized moving inputs use (3)–(4) independently, deliberately overestimating their joint range.

Proposition 2 (Interval result). *For $s = -1, +1$, every admissible normalized input, and $0 \leq \delta \leq 261/1024$,*

$$\operatorname{Re} X_s \geq 0.0041922727090, \quad \operatorname{Re} Z_s \geq 0.0655826074416.$$

The independently computed enclosures are:

	initial connector	final connector
minimum lower endpoint of $\operatorname{Re} X_s$	0.00419227270907	0.00419227270907
maximum upper endpoint of $\operatorname{Re} X_s$	0.218433271265	0.218433271265
minimum lower endpoint of $\operatorname{Re} Z_s$	0.0655826074417	0.0655826074417
maximum upper endpoint of $\operatorname{Re} Z_s$	0.776610992074	0.776610992074

The worst lower bound for X_s occurs in cell 159, at the outer collar boundary; the worst lower bound for Z_s occurs in cell 0, at the local endpoint. The two row lists agree exactly after conjugating the imaginary coordinate, so the checked one-side table in `research/chapter_vi_connector_seam_table.csv` represents both connectors. The verifier recomputes this symmetry rather than assuming it.

The calculation is reproduced by

```
python3 research/check_chapter_vi_connector_seam.py
```

The script uses only the Python standard library. Every elementary binary64 result is widened by one representable number toward each infinity. This is strong independent evidence and is close to a conventional interval proof, but it is not a kernel-checked certificate.

8 LeanCompCert formalization

The same calculation is now encoded independently in `ChapterVIDHomogeneousCompiledTable.lean`. Its primitive intervals and all intermediate rectangles have denominator 2^{20} . A row is not merely a list of final decimal endpoints: it is a straight-line trace containing the checked additions, multiplications, reciprocals, powers, and the rigorous exponential-remainder enclosure that produce those endpoints. Lean proves by structural interval lemmas that soundness of these operations implies containment of the literal complex expressions in (6)–(7).

There are 160 distance cells on each side. Ten rows, together with a 39-operation fixed prelude, form one artifact with 2609 dyadic operations and 23197 signed-integer claims. Thus the complete table consists of 32 independently checkable artifacts and 320 semantic rows. Generated Lean theorems, evaluated by the kernel rather than a native-decision shortcut, prove that every artifact lies inside LeanCompCert’s signed 64-bit fragment and that every one of its 23197 signed claims holds. LeanCompCert’s verified denotation theorem therefore proves that every reference artifact returns zero. The independent reference evaluation gives

quantity	worst precision-20 numerator
lower bound for $\text{Re } X_s$	3638
lower bound for $\text{Re } Z_s$	68596
upper bound for the exponential argument’s ℓ^1 norm	250724
failed signed-integer claims	0

Hence the formal table proves the slightly more conservative but still strict bounds

$$\text{Re } X_s \geq \frac{3638}{2^{20}} > 0, \quad \text{Re } Z_s \geq \frac{68596}{2^{20}} > 0, \quad \|a\|_1 \leq \frac{250724}{2^{20}} < 1.$$

The reference verdict is thus unconditional: it is derived from kernel proofs of the claims and the verified program semantics, not from an empirical execution premise. From that verdict, ordinary Lean theorems prove the two coefficient inequalities, the oriented derivative signs, the two crossing certificates, and finally the seam-compatible five-piece logarithmic limit. LeanCompCert can also derive emitted C from the exact operation lists; a hash-bound receipt interface is retained for optional independently compiled reruns. Neither such a receipt nor the Python table is used by the reference Lean theorem.

The reference computation is reproduced by

```
lake exe chapter-vi-connector-cert reference-homogeneous-shards
```

and reports 32 checked shards, 320 checked cells, and zero failed claims.

9 Conclusion for the seam

Theorem 3 (Oriented derivative on both collars). *For a sufficiently small anchored Morse model satisfying (3)–(4),*

$$-\frac{d}{dt} \text{Re } F(\zeta, u_-(t)) > 0 \quad \text{on the initial collar,}$$

and

$$\frac{d}{dt} \text{Re } F(\zeta, u_+(t)) > 0 \quad \text{on the final collar.}$$

Proof. The oriented derivative is the left side of (8). The formal LeanCompCert bounds, $L > 0$, and $\delta^2 \geq 0$ give

$$L \operatorname{Re} X_s + \delta^2 \operatorname{Re} Z_s \geq \frac{3638}{2^{20}} L + \frac{68596}{2^{20}} \delta^2 > 0.$$

Proposition 2 gives the stronger independent Python bounds quoted above. The orientation reverses the initial connector and leaves the final connector unchanged. $\square$

At the local endpoint, the real part of the first factor is already strictly positive: the product radicand is positive real there and the companion factor has positive real part. On the initial connector, Theorem 3 says that the first factor's real part is nonincreasing as t approaches 1, so every earlier collar value is at least its positive endpoint value. On the final connector it is nondecreasing away from $t = 0$. Consequently

$$\operatorname{Re} F(\zeta, u_s(t)) > 0 \quad \text{throughout both endpoint collars.}$$

In particular, if the imaginary part of the first factor vanishes at its unique possible crossing, its real part is positive. The principal square root therefore extends from the outer boundary to the prescribed local Morse root without crossing its branch cut. The existing positive-real result for the companion factor supplies the other square root. Their product gives the compatible radicand sheet on each connector, and the two local seam signs are the required positive Morse signs. Together with the already constructed outer arcs and pinched middle arc, this supplies the seam-compatible five-piece logarithmic contribution.

10 What the certificate actually computes

The certificate is needed because the endpoint displacement tends to zero with the selected Morse length. A direct rectangular enclosure of the unscaled derivative forgets that several small quantities are the same quantity. It consequently replaces exact cancellation by an interval containing both signs. The certificate does not establish a new abstract continuity principle. It verifies the finite uniform inequalities that remain after the analytic proof has retained the common scale symbolically.

For each side and distance cell, the Lean definition constructs the following dependency graph:

$$\begin{array}{c} (D, y_D, c, b, 10001^{-1}), \quad L, \delta, q_s, r_s, p \\ \downarrow \\ L^2 r_s, L q_s, V_s = B_s + L^2 r_s - s L q_s, u = D + s L q_s + \delta V_s \\ \downarrow \\ u^{-1}, u^{-2}, u^{-3}, u^{-4}, H(u), M(u), a = (u - D)H(u) \\ \downarrow \\ E(a), P(u), K(u) \\ \downarrow \\ X_s, Z_s, \operatorname{Re} X_s > 0, \operatorname{Re} Z_s > 0. \end{array}$$

Every arrow is represented by primitive dyadic interval operations. Products check all four endpoint products with outward rounding. Positive reciprocals first certify that the input interval is strictly positive and then certify both reciprocal endpoints. Complex products are four real products followed by exact signed additions. Powers and the removable inverse-power differences in Appendix A are built from these primitives; no floating-point transcendental function occurs in the artifact.

The relative exponential is treated analytically. Lean proves

$$e^a = 1 + a + a^2 E(a), \quad \|a\| \leq 1 \implies \|E(a)\| \leq 1,$$

and the table checks the stronger raw bound $\|a\|_1 \leq 250724/2^{20} < 1$. It then encloses the unknown complex value $E(a)$ by the unit square. Thus the finite computation only performs rational interval arithmetic; the transcendental estimate is an ordinary analytic theorem.

The fixed prelude contains 39 operations for the collision constants and base coefficients. Each cell contributes 257 further operations. One ten-cell shard therefore has

$$39 + 10 \cdot 257 = 2609$$

operations. Flattening their interval obligations produces 23197 signed product comparisons. There are 16 shards per side, so the full formal cover contains 32 artifacts and 320 cells. The fixed prelude is repeated in each shard to make every artifact independently checkable.

11 From integer comparisons to the analytic theorem

The formal proof has five distinct layers.

1. *Primitive containment.* Lean proves that the actual constants and normalized moving inputs lie in the declared dyadic boxes. In particular the inverse-Morse derivative estimate and the selector supply the box for q_s , rather than treating it as numerical input.
2. *Word-size admissibility.* For each shard the generated theorem `shard_admissible_SIDE_INDEX` proves in the kernel that every signed intermediate used by `LeanCompCert` fits the verified 64-bit fragment.
3. *Arithmetic truth.* The companion theorem `shard_allHold_SIDE_INDEX` proves all 23197 signed comparisons by kernel reduction. `LeanCompCert`'s verified program-denotation theorem converts admissibility plus these comparisons into the unconditional theorem that the corresponding checker program returns zero.
4. *Interval semantics.* The generic multiplication, reciprocal, and complex-trace lemmas turn the zero verdict into containment of every literal real or complex expression. The theorem `CellTrace.coefficients_pos_of_allSound` extracts the strict signs of $\text{Re } X_s$ and $\text{Re } Z_s$ for an arbitrary point covered by a row.
5. *Calculus and sheet assembly.* Lean identifies the row's coordinate with the literal affine connector coordinate, invokes the exact identity (8), and constructs an oriented-real derivative certificate on each collar. Existing one-variable calculus turns these into the two positive crossing certificates. The already certified companion factor, outer arcs, and connector regularity then construct compatible square-root sheets and the five-piece logarithmic limit.

The exported reference-model existence theorem has no numerical run, receipt, floating-point, or CSV premise. Its axiom audit contains only the standard mathlib foundational axioms `propext`, `Classical.choice`, and `Quot.sound`.

The same construction has now been applied to every older finite table consumed by the seam assembly: the 56 outer-polar shards, 64 connector-factor bulk shards, two endpoint anchors, 41

first-derivative shards, and 10 second-derivative shards. For each shard Lean proves both word-size admissibility and all signed comparisons, then obtains the zero-return theorem from Lean-CompCert’s verified semantics. The reference run verdicts therefore have no external execution premise. The newer theorem `radicand_ne_zero_on_puncturedRectangle` combines those verdicts with the homogeneous collar argument and discharges nonvanishing at every positive critical value. Its square-root sheet lives on the natural half-open parameter rectangle. The near- D gluing to the collision sheet and both outer sheets is now formalized; joining that terminal family to the original source cycle is analytic topology, not a missing execution receipt.

One further bookkeeping gap has been removed in Lean: `connectorIntegral_eq_principal_curveIntegral` constructs the actual source-coordinate connector path and proves the exact pull-back identity, including both the affine velocity and Poincaré’s nonlinear $u \mapsto t$ Jacobian. The former second interface, `principalPhiAlongCriticalValue`, has deliberately been removed. It re-evaluated the literal unit circle at the current parameter and therefore named the wrong analytic-continuation object. Theorems in `ChapterVIDRadialContour.lean` prove that the outside collision pole eventually lies inside that circle, and `ChapterVIWindingObstruction.lean` proves that no pole-avoiding continuation can preserve the incompatible winding numbers. The missing object is a lift of the original germ through the long parameter range before the certified near- D jointly moving cycle begins.

For §102, the primary text on pp. 326–329 gives a staged parameter count rather than assuming a two-coordinate factorization at the outset. The new theorem `collisionOrientation_rank_le_two_of_ratio_rank_le_four` formalizes that count for an arbitrary family of collision roots. The theorem `not_fourParameterRatioFactorization` then connects the resulting rank bound directly to the 24 certified moving branches of §103. The same file now formalizes the preceding Chapter V calculation rather than assuming (13 bis): it derives equation (13) from the first-order Fourier coefficient identity and resonance, proves the corrected $H = -\zeta S$ cancellation, proves the characteristic direction nonzero from independence, and obtains rank five by rank-nullity. The exact theorem `chapterVNo85_printed_normalization_counterexample` records why the printed $-\zeta H = S$ cannot be used literally.

12 The terminal lifted contour and Chapter V extraction

The finite tables now feed an actual two-dimensional continuation object rather than five unrelated paths. Let

$$P = [0, 1) \times [0, 1],$$

where the first coordinate is the positive critical-value parameter and the second traverses the closed contour. The function `positiveContourPoint` concatenates, in order, the initial outer quarter, upper connector, inverse-Morse middle segment, lower connector, and final outer quarter. Nested path concatenation is reparametrized exactly by

$$[0, \tfrac{1}{4}], \quad [\tfrac{1}{4}, \tfrac{1}{2}], \quad [\tfrac{1}{2}, \tfrac{5}{8}], \quad [\tfrac{5}{8}, \tfrac{3}{4}], \quad [\tfrac{3}{4}, 1].$$

The endpoint equalities are proved in the path types, so this is a closed path for every parameter and not an algebraic formal sum of integrals.

Write $R(s, t)$ for Poincaré’s literal transformed source radicand evaluated on this path. Lean proves

$$R(s, t) \neq 0 \quad ((s, t) \in P)$$

by dispatching each of the five subintervals to the compiled outer tables, the full connector tables, or the exact identity $R = k + v^2$ on the positive Morse segment. Continuity and the covering-space lift for $w \mapsto w^2$ produce a continuous $W : P \rightarrow \mathbb{C}$ with

$$W(s, t)^2 = R(s, t).$$

It is normalized to the principal root at one basepoint. Connectedness and square-root uniqueness then propagate that normalization across every seam. The theorem `exists_jointSheet_eq_allCertifiedPieces` states simultaneously that W restricts to the two compiled principal outer roots, both seam-compatible connector roots, and the positive Morse root. Thus the entire terminal five-piece branch choice is kernel checked. This theorem starts inside the certified near- D neighborhood; it does not invent the unprinted long-range homotopy from the original §94 circle.

The Chapter V layer is likewise now an analytic extraction rather than six unrelated scalar assumptions. On Mathlib’s two-dimensional unit additive torus, suppose one has the pointwise first-order identity

$$-S \partial_r B + T \partial_r C + d_X B = 0.$$

Linearity of Haar integration proves that its Fourier coefficient at a selected mode λ satisfies

$$-S \lambda B_\lambda + T \lambda C_\lambda + dB_\lambda(X) = 0,$$

once the standard Fourier multiplier and differentiation-under-the-integral identities are supplied. Applying this one theorem to a head mode and five tail modes constructs all six fields of the formal resonant first-order data structure. Resonance $T = 0$, independence, and the corrected normalization $H = -\zeta S$ then give the common characteristic direction and the rank chain

$$6 \longrightarrow 5 \longrightarrow 4 \longrightarrow 2,$$

which the certified §103 calculation contradicts. What is still source-specific is deriving the multiplier, interchange, resonance, and independence hypotheses from the full Hamiltonian and putative uniform integral, and proving that these six Fourier observables are exactly the common-radius contour-singularity observables.

13 Independent checks and reproducibility

The kernel proof and the compact Python computation intentionally use different implementations. They agree on the sign and have comfortably positive margins; the Python implementation obtains the stronger enclosures in Proposition 2, while the formal table obtains the more conservative dyadic bounds used in Theorem 3. The following commands reproduce the checks from the repository root:

```
python3 research/check_chapter_vi_connector_seam.py
lake build PoincareChapterVI.ChapterVIDHomogeneousAdmissibility
python3 research/generate_radial_tail_kernel_shards.py
lake build PoincareChapterVI.ChapterVIDRadialTailFinal
lake exe chapter-vi-connector-cert reference-homogeneous-shards
lake env lean PoincareChapterVI/AxiomAudit.lean
lake build
```

The observed reference summary is


```

checked homogeneous shards: 32
checked homogeneous cells: 320
failed integer claims: 0
minimum endpoint-coefficient lower numerator / 2^20: 3638
minimum distance-coefficient lower numerator / 2^20: 68596
maximum exponential-argument L1 numerator / 2^20: 250724

```

The executable can also emit any shard as straight-line C and submit it to CompCert:

```

lake exe chapter-vi-connector-cert \
  emit-homogeneous-shard initial 0 shard.c
lake exe chapter-vi-connector-cert \
  check-homogeneous-shard initial 0

```

The second command requires the CompCert driver `ccomp` on `PATH`. It is an optional independently compiled rerun; it is not required for the unconditional kernel theorem.

14 Precise project status

This work closes the local connector-seam obstruction at the certified point D and the former final radial-tail obstruction. It also shows that the two obligations previously described as literal source transcription cannot both be proved as stated: the fixed-current-unit-circle continuation is topologically impossible, and the normalization printed in Chapter V no. 85 is algebraically false. The corrected finite no. 85 calculation is now formalized. Poincaré’s complete Chapter VI proof is not thereby complete.

Two corrected source-reconstruction obligations remain outside the scope of the finite certificate:

1. In §§95–100, one must lift the original §94 germ along a *jointly moving* closed source cycle that avoids every moving radicand zero until it enters the certified near- D family, and control the complementary regular arcs along that transport. It is not enough to deform the unit circle re-evaluated at the terminal parameter: Lean proves that the relevant pole has changed sides of that fixed circle. The terminal five-piece contour, its positive-parameter square-root sheet, and the identification of all five branches are closed; what is missing is the earlier global joint cycle and sheet transport.
2. In §102, one must derive the recorded Fourier multiplier, orbital-interchange, resonance, and independence fields from the full source Hamiltonian and Chapter V uniform integral, and identify the six rescaled coefficients with the common-radius singularity data of the continued contour. Once supplied, the corrected (13 bis) construction, five-to-four-to-two rank chain, Darboux transfer, and §103 contradiction apply without another finite-algebra premise.

These are analytic source-reconstruction problems, not further rows missing from the interval table. Enlarging the present certificate would not discharge them.

15 Formal source map

The main files supporting this note are:

file	role
ChapterVIDHomogeneousDerivative.lean	exact homogeneous cancellation and semantic coefficient certificate
ChapterVIDHomogeneousCompiledTable.lean	dyadic traces, cell cover, semantic soundness, and seam theorem bridge
ChapterVIDHomogeneousAdmissibility.lean	aggregate unconditional reference verdict and exported model-existence theorem
ChapterVIDConnectorFullBulk.lean	uniform two-dimensional reuse of the bulk table, positive-fiber collar closure, and continuous square-root sheets on the two half-open connector rectangles
ChapterVIDJointLiftedContour.lean	one closed terminal five-piece source contour, literal-radical and nonvanishing on its positive half-open parameter family, a basepoint-normalized joint square-root sheet, and exact identification with all five certified branches
ChapterVIDRadialTailCenteredCertificate.lean	14,336-cell centered affine/Taylor certificate and semantic proof that the literal radial derivative has negative real part
ChapterVIDRadialTailEndpointCertificate.lean	direct endpoint table, exact double-zero anchors, positive-curvature collars, and endpoint nonnegativity
ChapterVIDRadialTailFinal.lean	unconditional full-tail monotonicity certificate, complete pre-collision real-part positivity, and principal square-root sheet
RadialTailCenteredKernel/*.lean, RadialTailEndpointKernel/*.lean generate_radial_tail_kernel_shards.py	generated, independently buildable decide + kernel shards for every finite table assertion deterministic generator for the centered and endpoint kernel modules and their aggregate theorems
ChapterVIWindingObstruction.lean, ChapterVIDRadialContour.lean	winding invariance, the concrete outside-to-inside crossing, and impossibility of the former fixed-unit-circle continuation premise
ChapterVISection102.lean	pre-resonance no. 85 Fourier equation, exact counterexample to the printed normalization, corrected equation-(13 bis) kernel construction, and complete rank chain from six scaled values to the §103 contradiction
ChapterVNo85FourierExtraction.lean	actual two-torus Haar Fourier extraction of the head and five tail equations from one pointwise first-order Poisson identity
HomogeneousAdmissibility/*.lean	independently buildable kernel proofs for all 32 shards
ConnectorCertificateMain.lean	reference, statistics, emission, and native-check commands
check_chapter_vi_connector_seam.py chapter_vi_connector_seam_table.csv generate_chapter_vi_homogeneous_admissibility.py	independent outward-rounded Python audit compact symmetric Python table 18 reproducible generator for the kernel shard modules

A Explicit coefficient formulas

Put

$$c = \frac{100}{30003}, \quad b = \frac{200}{10001}, \quad Q = u^2 + uD + D^2, \quad Q_1 = u + 2D.$$

The coefficient of $u - D$ in the relative exponential argument and the multiplier are

$$\begin{aligned} H(u) &= -cQ(1 + u^{-3}D^{-3}), \\ M(u) &= -2y_D D^{-1} + by_D D^{-1}(u^{-3} + u^3). \end{aligned}$$

The argument in (2) is $a = (u - D)H(u)$, and

$$P(u) = (1 + a + a^2 E(a))M(u).$$

For the quadratic coefficient, introduce

$$\begin{aligned} \eta_2 &= -\frac{u + D}{u^2 D^2}, & \eta_4 &= -\frac{u^3 + u^2 D + u D^2 + D^3}{u^4 D^4}, \\ T &= D^{-4}(u^{-2} + D^2 u^{-4}), & T_1 &= D^{-4}(\eta_2 + D^2 \eta_4), \\ R_L(u) &= \frac{1}{10001}((30000 + 3T) + 6DT_1), \\ \eta_3 &= -\frac{Q}{u^3 D^3}, \\ R_S(u) &= D^{-1}[Q_1(1 - u^{-3}D^{-3}) + 3D^2(-D^{-3}\eta_3)], \\ H_1(u) &= -c[Q_1(1 + u^{-3}D^{-3}) + 3D^2(D^{-3}\eta_3)], \\ M_1(u) &= by_D D^{-1}(\eta_3 + Q). \end{aligned}$$

Then

$$K(u) = R_L(u) + by_D R_S(u) + H_1(u)M(u) + H(D)M_1(u) + H(u)^2 E(a)M(u).$$

Finally, define

$$\begin{aligned} C_L(u) &= \frac{u + D}{10001} \left(30000 + 3 \frac{u^2 + D^2}{u^4 D^4} \right), \\ C_S(u) &= QD^{-1}(1 - u^{-3}D^{-3}), \\ C(u) &= C_L(u) + by_D C_S(u). \end{aligned}$$

The base coefficient is evaluated without numerical differentiation:

$$A = C(D) + H(D)M(D), \quad -2.69848482771 \leq A \leq -2.56120107040.$$